

PVsyst - Simulation report

Grid-Connected System

Project: EC9180_PVsyst_Lab2.

Variant: New simulation variant

Sheds, single array

System power: 105 kWp

Kilinochchi - Sri Lanka

**PVsyst V8.1.5**

VC0, Simulation date:
14/08/26 16:02
with V8.1.5

Project summary**Geographical Site****Kilinochchi**

Sri Lanka

Situation

Latitude 9.3840 °(N)

Longitude 80.4087 °(E)

Altitude 29 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

Kilinochchi

Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary**Grid-Connected System****Orientation #1****Fixed plane**

Tilt/Azimuth 9.5 / 0 °

System information**PV Array**

Nb. of modules

190 units

Pnom total

105 kWp

Sheds, single array**Near Shadings**

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

Inverters

Nb. of units

1 unit

Total power

100 kWac

Pnom ratio

1.05

Results summary

Simulation step Hourly

Produced Energy 166475 kWh/year Specific production 1593 kWh/kWp/year Perf. Ratio PR 82.86 %

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General parameters**Grid-Connected System****Sheds, single array****Simulation step**

Hourly

Orientation #1**Fixed plane**

Tilt/Azimuth 9.5 / 0 °

Sheds configuration

Nb. of sheds 5 units

Single array

Shading limit angle

Limit profile angle 59.6 °

Sizes

Sheds spacing 2.50 m

Sensitive width 2.29 m

GCR Shading 92.3 %

Top inactive band 0.02 m

Bottom inactive band 0.02 m

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

PV Array Characteristics**PV module**

Manufacturer Generic

Model Mono 550 Wp Twin half-cells bifacial
(Original PVsyst database)

Unit Nom. Power 550 Wp
Number of PV modules 190 units
Nominal (STC) 105 kWp
Modules 10 units x 19 In series

At operating cond. (50°C)

Pmpp 95.9 kWp
U mpp 718 V
I mpp 134 A

Total PV power

Nominal (STC) 105 kWp
Total 190 modules
Module area 491 m²
Cell area 453 m²

Inverter

Manufacturer Generic

Model 100 kWac inverter
(Original PVsyst database)

Unit Nom. Power 100 kWac
Number of inverters 2 * MPPT 50% 1 unit
Total power 100 kWac
Operating voltage 180-1000 V
Pnom ratio (DC:AC) 1.05
No power sharing between MPPTs

Total inverter power

Total power 100 kWac
Number of inverters 1 unit
Pnom ratio 1.05

Array losses**Thermal Loss factor**

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 88 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.38 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

Strings Mismatch loss

Loss Fraction 0.10 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

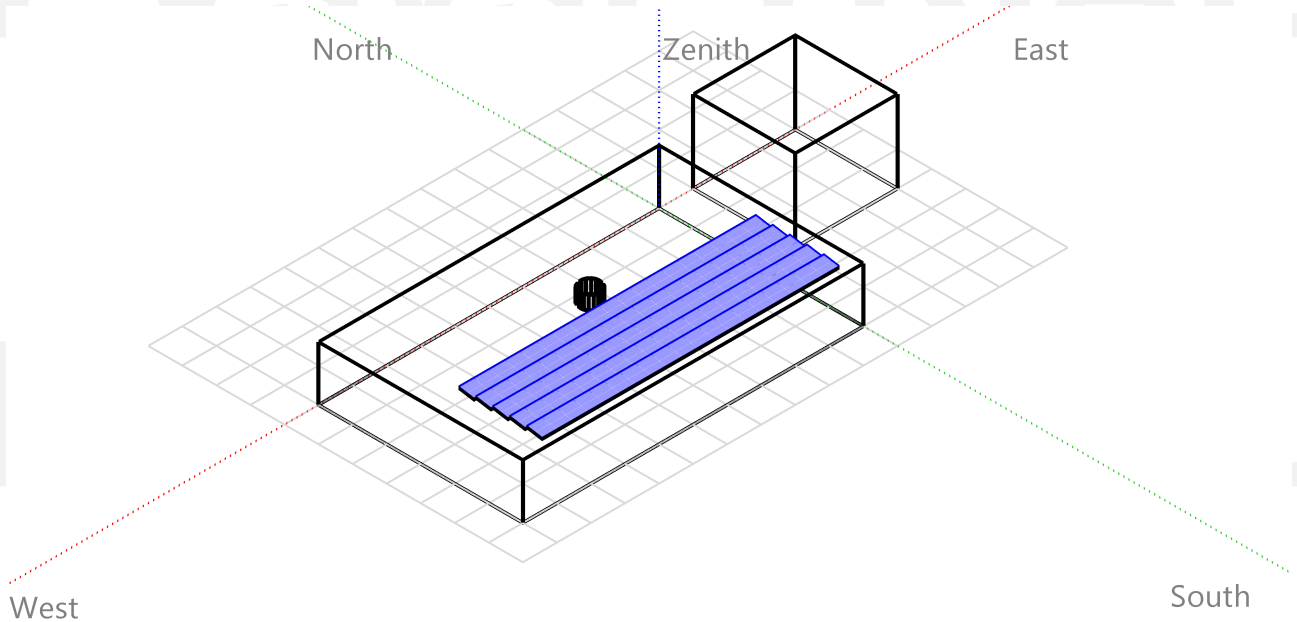


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Near shadings parameter

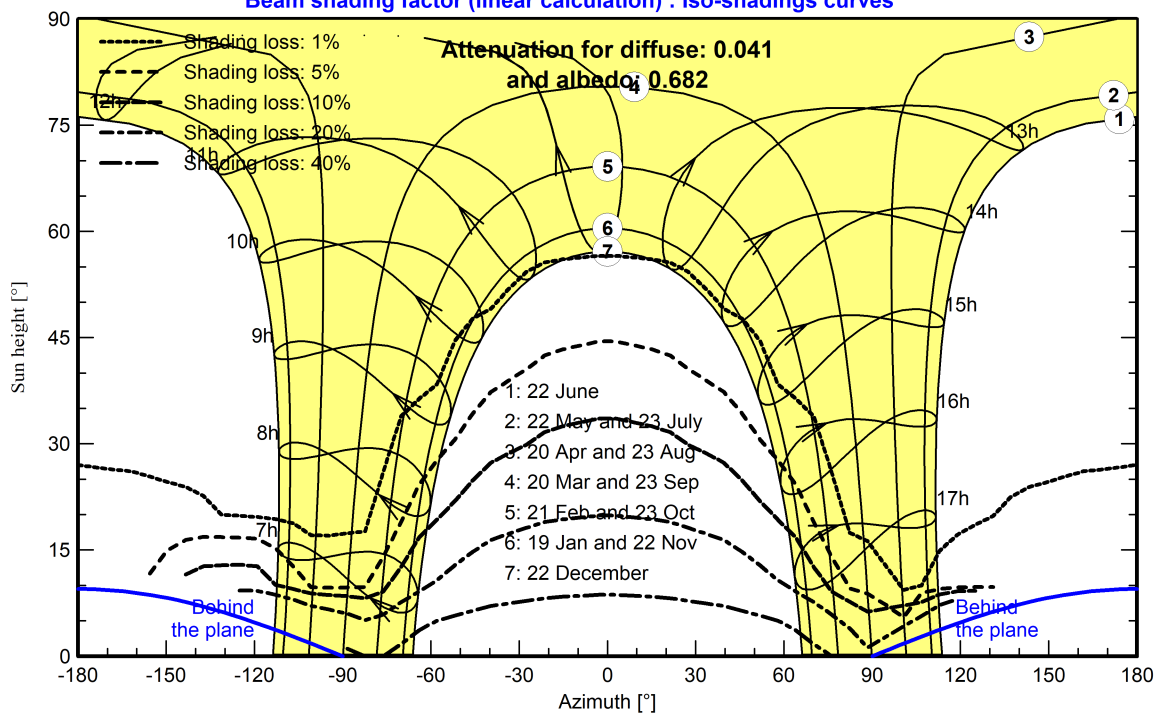
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 9.5°/ 0°

Beam shading factor (linear calculation) : Iso-shadings curves





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Main results

Simulation step

Hourly

System Production

Produced Energy

166475 kWh/year

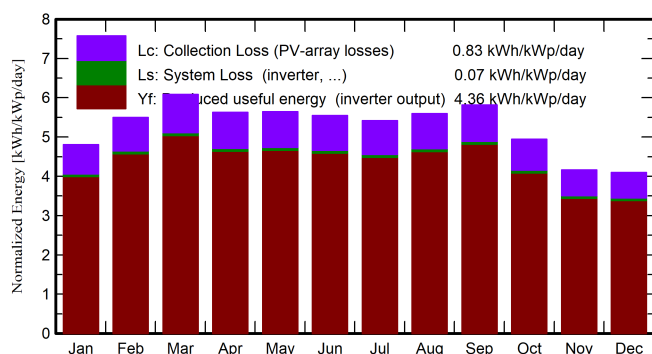
Specific production

1593 kWh/kWp/year

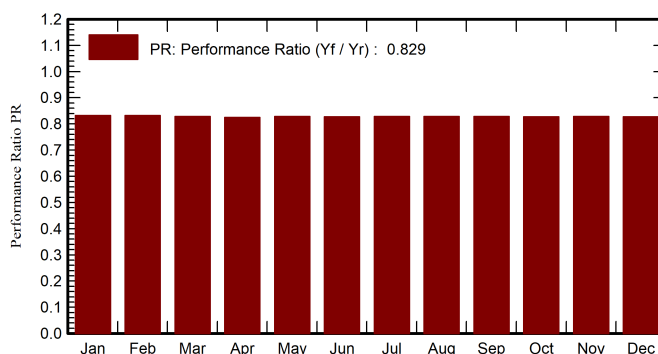
Perf. Ratio PR

82.86 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m²	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	ratio
January	138.4	67.23	25.56	149.0	141.9	13175	12955	0.832
February	146.2	60.86	26.51	154.0	148.7	13605	13391	0.832
March	184.8	78.57	28.57	188.6	182.8	16579	16331	0.829
April	171.9	81.15	30.10	168.9	163.0	14775	14547	0.824
May	182.2	89.05	30.32	174.9	168.8	15369	15135	0.828
June	176.0	80.75	30.28	166.5	160.4	14635	14400	0.828
July	176.3	86.42	29.86	167.9	161.7	14789	14548	0.829
August	177.9	83.32	29.34	173.4	167.5	15261	15018	0.829
September	173.1	77.08	28.88	174.4	168.8	15339	15108	0.829
October	147.7	71.46	27.95	153.2	147.6	13477	13253	0.828
November	117.8	67.10	26.72	124.8	118.7	11011	10812	0.829
December	118.1	64.60	25.79	126.9	119.9	11183	10977	0.828
Year	1910.5	907.58	28.33	1922.5	1849.8	169199	166475	0.829

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

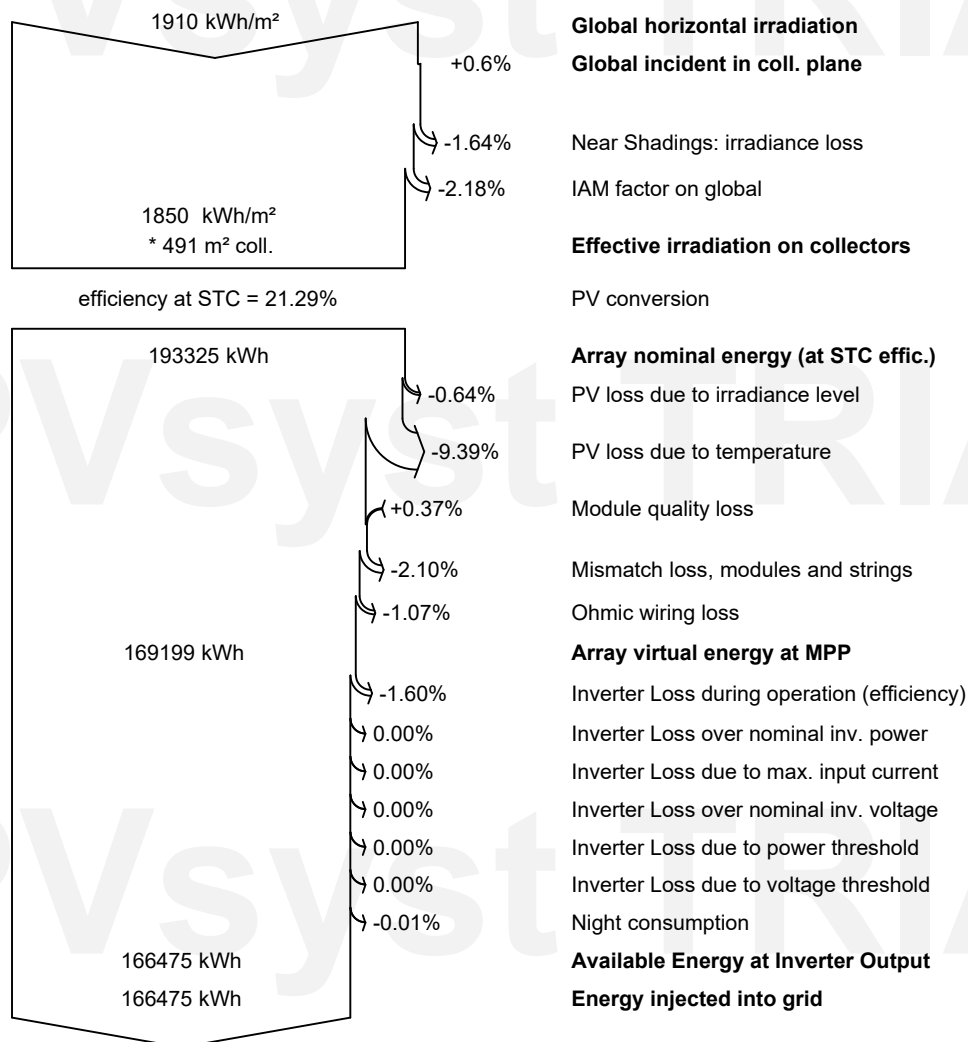
PR Performance Ratio



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Loss diagram

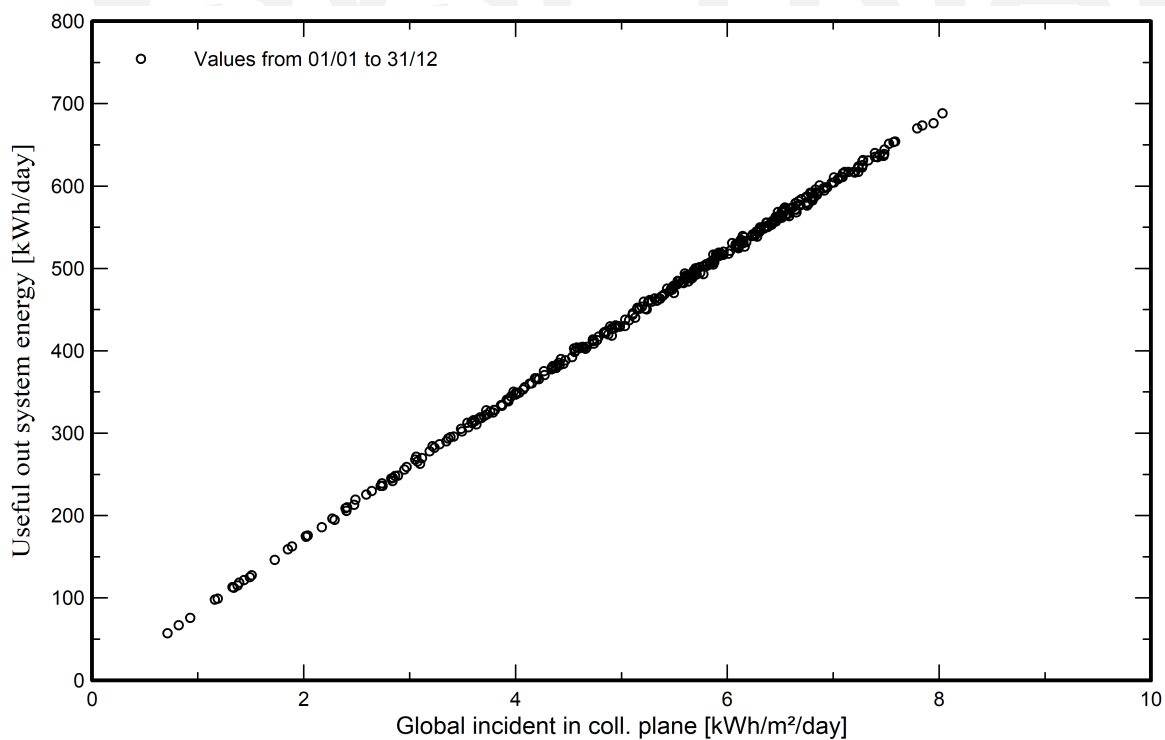




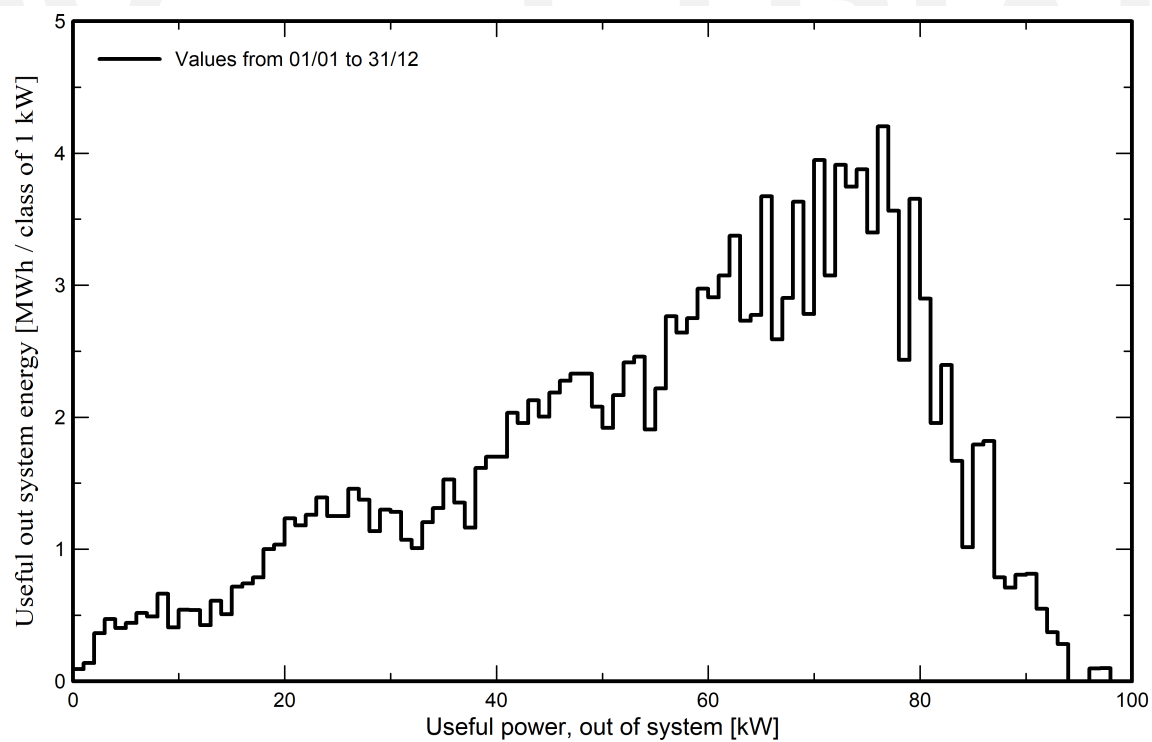
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Predef. graphs
Daily Input/Output diagram



System Output Power Distribution

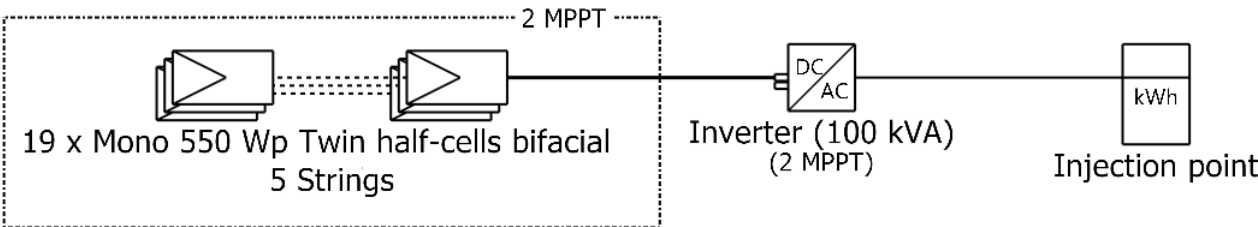




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Single-line diagram



PV module	Mono 550 Wp Twin half-cells bifacial
Inverter	100 kWac inverter
String	19 x Mono 550 Wp Twin half-cells bifacial

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